

Machine Calculation Sheets — Zone 1.2

WANKOE limestone processing line — independent design-verification document

NOEZYS · AI Innovation Lab — working version, 2026-08-12 — confidential — companion to "Machine Calculation Sheets — Zone 1"

Scope and method

This document presents the engineering calculation of the machines of Zone 1.2 of the WANKOE limestone processing line, for independent validation. It follows the same method as the Zone-1 companion document: for each machine, its function, input data, constraints, the engineering model with every symbol defined and its literature source, the numerical application on validated project data, and the computed results. All figures come from the same deterministic flowsheet model (closed mass balances, closed circuits solved by fixed-point iteration on the recirculating stream). The document is self-contained.

Zone 1.2 in the plant

Zone 1 (companion document) produces a **0/20 mm stockpile** alongside the KFS kiln stone. Zone 1.2 reclaims this stockpile and converts it into two products: **AgLime 0/1.7 mm** (agricultural lime, market cap 135 000 t/y) and **FeedLime 5–15 mm**, the feedstock of Zone 1.3 (drying/milling for feed-grade products, whose firm 40 000 t/y grits contract creates a FeedLime demand of 162 075 t/y). The chain: reclaim feeder **BF.5101** (100 t/h, no process calculation — a constant-rate extractor) → double-deck screen **SR.5105** (15/5 mm) → the 5–15 mm mid cut goes to the FeedLime stockpile, while the +15 and 0–5 cuts feed the **AgLime closed loop**: closing screen **SR.5115** (1.7 mm) whose oversize is crushed by impactor **CR.5107** and returned.

Operating modes. **2A** (computed here): mid cut → FeedLime, rest → AgLime loop. **2B**: the whole reclaim → FeedLime, loop stopped — this needs no machine model: it is the exact mass identity FeedLime = reclaim. Rain forces 2B, because the 1.7 mm cut is physically impossible on wet fines (the partition collapses); 2B is also selectable in dry weather. **2C**: everything loops back to stock (flushing/return mode, no product; not computed).

Design basis (client-fixed, 2026-08-12)

Item	Value	Origin / status
Reclaim rate (BF.5101)	100 t/h wet = 93.0 t/h dry	client decision
Reclaimed 0/20 PSD	Zone-1 product curve at validated settings (P_{80} = 13.66 mm)	client decision — consistent with the companion document
Reclaim moisture	7 % (wet basis)	declared hypothesis: stockpile moisture = belt-measured value
Weather	dry photo computed; rain = forced mode 2B (physical rule); season split 75 % dry / 25 % rain (39/13 weeks)	specification rule

Planning driver	Zone-1.3 FeedLime demand (162 075 t/y), AgLime produced up to its 135 kt/y market cap — zero unsellable surplus	client decision
AgLime acceptance criterion	$\geq 95\%$ passing 1.7 mm	client-adopted project criterion (no prior contractual spec)
Screen imperfection I	0.15 (dry) $\rightarrow$ sharpness $s = 13.52$; rain: cut impossible	project arbitration; 0.10 variant under expert review
Bond work index / efficiency	$W_i = 12.54 \text{ kWh/t} \cdot \eta_m = 0.75$	[ref.] / project
Shift regime	$3 \times 8\text{-h}$ shifts, 21 shifts/week = 168 h, maintenance reserve 23.8 h/week $\rightarrow$ net ceiling 144.2 h/week (7 500 h/y)	client operating frame

Sheet 1 · SR.5105 — double-deck reclaim screen 15/5 mm

Primary split of the reclaimed 0/20 · feed: BF.5101 · products: +15 and 0–5 (to AgLime loop) · 5–15 = FeedLime

1.1 Function

SR.5105 performs the three-way split that separates the FeedLime product from the AgLime loop feed. Its top deck (15 mm) removes the coarse fraction; its bottom deck (5 mm) isolates the 5–15 mm FeedLime cut from the 0–5 mm fines. Routing depends on the operating mode (§ preamble); in mode 2A the +15 and 0–5 streams merge into the closed loop.

1.2 Input data

Quantity	Value	Origin
Throughput (dry)	93.0 t/h	100 t/h wet reclaim at 7 % moisture
Feed P_{80}	13.66 mm	zone-1 0/20 product curve

1.3 Model — partition curve (Karra)

A real screen deck gives a particle of size x a probability $\rho(x)$ of reporting to the oversize; Karra (1979) [3] describes it by a log-logistic function centred on the corrected cut size:

$$\rho(x) = \frac{1}{1 + (d_{50c}/x)^s}, \quad d_{50c} = k_d a, \quad s = \frac{\ln 9}{\ln(1/(1 - I))} \quad (1.1)$$

where the sharpness s follows from the quartile definition $\rho(d_{25}) = 0.25$, $\rho(d_{75}) = 0.75$ and the imperfection I is defined by $d_{25}/d_{75} = 1 - I$ (derivation in the companion document, Sheet 2). With $I = 0.15$: $s = 13.52$. Flows are computed class by class and mass is conserved exactly. Required deck area follows the VSMA/Fontaine capacity law:

$$Q_b = 14 a^{0.6} \text{ [t/h/m}^2\text{]}, \quad A = \frac{U f_p}{Q_b f_0}, \quad f_p = 1.2, \quad f_0 = 0.347 \quad (1.2)$$

The capacity law is a field heuristic flagged for cross-checking (companion document, §2.5 note). Under rain, wet screening loses 25 % capacity — but rain forces mode 2B on this zone anyway (the sizing consequence is shown in §1.5).

1.4 Numerical application

$$Q_{b,\text{top}} = 14 \times 15^{0.6} = 71.1, \quad A_{\text{top}} = \frac{78.25 \times 1.2}{71.1 \times 0.347} = 3.81 \text{ m}^2 \quad (1.3)$$

$$Q_{b,\text{bot}} = 14 \times 5^{0.6} = 36.8, \quad A_{\text{bot}} = \frac{31.25 \times 1.2}{36.8 \times 0.347} = 2.94 \text{ m}^2 \quad (1.4)$$

1.5 Results

Stream	Dry (t/h)	Wet (t/h)	P_{80} (mm)	Destination (mode 2A)
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+15 oversize	14.75	15.86	19.44	AgLime loop
5–15 = FeedLime	47.00	50.54	12.23	FeedLime stockpile (Zone-1.3 feed)
0–5 undersize	31.25	33.60	4.14	AgLime loop

Sieve (mm)	0.5	1	2	4	5	8	10	12.5	15	20
Feed 0/20	1.8	4.6	11.2	26.0	33.3	53.0	63.9	75.4	84.9	98.0
+15	0	0	0	0	0	0	0.1	1.4	14.9	87.2
FeedLime 5–15	0	0	0	0.1	2.7	38.4	60.0	82.2	96.7	100
0–5	5.4	13.6	33.4	77.2	95.1	100	100	100	100	100

Required areas: top 3.81 m², bottom 2.94 m² (dry). **The sizing case is rain, mode 2B** (screen still runs, capacity × 0.75): 5.08 / 3.92 m².

Vendor data gap. Installed deck areas unknown; requirement to state to the vendor: 5.08 / 3.92 m² (rain case). BF.5101 feeder datasheet also not provided.

Sheet 2 · SR.5115 — closing screen 1.7 mm (AgLime loop)

Closes the AgLime loop · feed: +15 and 0–5 from SR.5105 + CR.5107 recycle · undersize 0/1.7 = AgLime product

2.1 Function

SR.5115 makes the AgLime product cut. Its oversize goes to the impact crusher CR.5107 and returns: screen and crusher form a closed circuit. A structural property of any closed loop follows from mass conservation at stationarity: **everything that enters the loop leaves as undersize** — the AgLime rate equals the fresh loop feed, independently of the crusher setting, which only governs the circulating load:

$$\dot{m}_{\text{AgLime}} = \dot{m}_{\text{loop,in}}, \quad CL = \frac{R}{F} = \frac{1 - e}{e} \quad (2.1)$$

R recirculated flow; F fresh loop feed; e screen extraction efficiency

2.2 Input data (converged loop state)

Quantity	Value	Origin
Fresh loop feed (+15 & 0–5)	46.00 t/h dry	SR.5105, mode 2A
Screen feed (fresh + recycle)	83.74 t/h dry	engine, converged
Screen feed P_{80}	4.91 mm	engine

2.3 Model and validity

Karra partition (1.1) at $a = 1.7$ mm, $I = 0.15$, $s = 13.52$; area law (1.2). *Validity note: Karra's model is established for apertures > 1 mm — the 1.7 mm cut sits near this limit; the partition is retained as the best available model, flagged for the ongoing expert review. Under rain the imperfection collapses and the cut becomes impossible: this is the physical basis of the forced mode-2B switch, not a modelling choice.*

2.4 Numerical application and results

$$Q_b = 14 \times 1.7^{0.6} = 19.2 \text{ t/h/m}^2, \quad A = \frac{46.0 \times 1.2}{19.2 \times 0.347} = 8.26 \text{ m}^2 \quad (2.2)$$

Stream	Dry (t/h)	Wet (t/h)	P_{80} (mm)	Destination
+1.7 oversize	37.74	40.58	17.17	CR.5107
0/1.7 = AgLime	46.00	49.46	1.065	AgLime product

Sieve (mm)	0.063	0.25	0.5	1.0	1.5	1.7	2
AgLime, % passing	3.0	19.0	42.5	77.2	95.4	98.8	99.9

AgLime quality — acceptance criterion ≥ 95 % passing 1.7 mm: achieved 98.8 % $\rightarrow$ compliant. The oversize P_{80} (17.2 mm) is coarse because the fresh +15 lumps concentrate in the oversize until crushed.

Circulating load: $37.74/46.00 = 82\%$ of fresh feed. Verification of (2.1): loop in 46.00 = AgLime out 46.00 t/h dry.

Vendor data gap. Installed area unknown; the 8.26 m² requirement makes SR.5115 the largest screen of the plant (fine apertures pass little flow per unit area).

Sheet 3 · CR.5107 — impact crusher (AgLime loop)

Crushes the +1.7 oversize and returns it to SR.5115 · the heaviest specific comminution duty of the line

3.1 Function and settings

CR.5107 closes the AgLime loop by crushing the +1.7 mm oversize to near-product size in one pass. Per (2.1) its settings do not change the AgLime rate; they set the circulating load and its own power. Settings: rotor speed $v = 40$ m/s, CSS = 1.0 mm — the minimum-circulating-load recommendation of the project's model book (CSS 1.0–1.2 mm, $v \approx 40$ m/s). Breakage parameters $A = 60$, $b = 0.80$ [H] (calcite drop-weight literature gives softer values, A 62–69, b 1.3–3.0 [6]; kept conservative pending site tests).

3.2 Model — impact fragmentation (JKMRC t_{10})

A particle struck at peripheral speed v receives the specific kinetic energy $\frac{1}{2}v^2$ (J/kg), i.e. in kWh/t (1 kWh/t = 3600 J/kg):

$$E_{cs} = \frac{v^2}{7200}, \quad t_{10} = A (1 - e^{-b E_{cs}}), \quad n = \max\left(0.65, \left(\frac{30}{t_{10}}\right)^{0.30}\right) \quad (3.1)$$

(Narayanan & Whiten, JKMRC [7]). The product follows the truncated Rosin–Rammler distribution with $x_{80} = \text{CSS}$ (companion document, Sheet 1, eq. 1.2), and power follows Bond (companion document, eq. 1.3). The loop is solved by fixed-point iteration until the recycle stream converges in flow and size distribution.

3.3 Numerical application

$$E_{cs} = \frac{40^2}{7200} = 0.2222 \text{ kWh/t}, \quad t_{10} = 60 (1 - e^{-0.8 \times 0.2222}) = 9.77 \%, \quad n = 1.400 \quad (3.2)$$

$$W = 10 \times 12.54 \left(\frac{1}{\sqrt{947}} - \frac{1}{\sqrt{17167}} \right) = 3.12 \text{ kWh/t} \quad (3.3)$$

$$P_{\text{net}} = 3.12 \times 37.74 = 117.7 \text{ kW}, \quad P_{\text{inst}} = \frac{117.7}{0.75} = 156.9 \text{ kW} \quad (3.4)$$

Validity: the reduction $17.2 \rightarrow 0.95$ mm (ratio 18) is far beyond Bond's comfort zone (ratios 3–6); $W = 3.12$ kWh/t is the line's heaviest specific duty and its least certain power figure — treat as an order-of-magnitude for motor selection, to be confirmed by the vendor.

3.4 Results

Sieve (mm)	0.125	0.25	0.5	1.0	1.5	1.7	2
Feed, % passing	0	0	0	0	0.3	2.0	5.9
Product, % passing	8.7	21.4	47.3	82.8	97.5	100	100

Throughput 37.74 t/h dry (converged) · product $P_{80} = 0.95$ mm, 97.5 % < 1.5 mm — one pass nearly closes the cut, which is why the circulating load stays at 82 %.

Vendor data gap. No motor rating provided; the computed requirement is 156.9 kW installed — the largest motor of zones 1–1.2, to be confirmed against vendor data given the Bond-validity caveat above.

4 · Zone balance, mode 2B, and weekly production plan

4.1 Mass balance (mode 2A, dry)

93.00 = 47.00 + 46.00 t/h dry (relative gap < 10⁻⁶) (4.1)

reclaim FeedLime AgLime

4.2 Mode 2B (rain, or by choice)

The loop stops; the mass identity FeedLime = reclaim holds exactly: 100 t/h wet of 0/20 becomes FeedLime at 100 t/h wet. No machine calculation is involved; SR.5105 runs as a bypass conveyor (its rain sizing is in §1.5). Mode 2B makes FeedLime **twice as fast** as mode 2A (100 vs 50.5 t/h wet) — the fact the weekly plan exploits.

4.3 Weekly production plan (client planning rule: FeedLime demand drives; AgLime up to its market cap, zero surplus)

Zone 1.3's FeedLime demand is 162 075 t/y = **3 117 t/week**, continuous. The AgLime market cap (135 000 t/y) limits mode-2A operation; the FeedLime balance is completed in mode 2B during the 13 rain weeks. Weekly ceiling: 21 × 8-h shifts = 168 h, of which 23.8 h/week are reserved for maintenance → net 144.2 h/week; hours below are clock hours at 80 % availability.

Week type	Count	Mode	Clock h/wk	FeedLime (t/wk)	AgLime (t/wk)	Ceiling use
Dry-season week	39	2A	87.5	3 537	3 461	61 %
Rain-season week	13	2B	23.2	1 855	0	16 %
Year	52	—	3 713 h/y	162 075	135 000	49.5 % of 7 500 h

The seasonal FeedLime imbalance is carried by the stockpile: dry weeks overproduce (+420 t/wk to stock), rain weeks underproduce (−1 262 t/wk from stock); the two balance over the year (39 × 420 ≈ 13 × 1 262). The stockpile must therefore hold a seasonal swing of ≈ **16 400 t** of FeedLime — a storage-capacity requirement for the yard design.

Planning statement. Zone 1.2 is comfortable: 49.5 % of its hour ceiling serves both the firm-grits-driven FeedLime demand and the full AgLime market, with zero unsellable surplus. Its feasibility is *conditional on Zone 1* delivering the 0/20 stockpile (26.5 kt/month in mode 1A at 250 t/h — companion document §4.2) and on the ≈ 16 400 t FeedLime seasonal storage.

5 · Hypothesis register and open points

#	Hypothesis	Value	Status
H1	Reclaimed 0/20 PSD	zone-1 photo curve at validated settings	client decision; no stockpile measurement will ever exist (project measurement policy)
H2	Stockpile moisture = belt moisture	7 %	declared hypothesis
H3	Screen imperfection (dry)	$I = 0.15 \rightarrow s = 13.52$	project arbitration; 0.10 under expert review (moves the split and circulating load)
H4	Karra validity at 1.7 mm	model retained at its validity limit	documented caveat; expert review pending
H5	Rain rule	1.7 mm cut impossible $\rightarrow$ mode 2B forced	specification physical rule
H6	Screen capacity law	$Q_b = 14a^{0.6}$, $f_0 = 0.347$, $f_p = 1.2$; rain $\times 0.75$	field heuristic, flagged for cross-check
H7	Impact breakage pair (CR.5107)	$A = 60$, $b = 0.80$	[H] conservative; drop-weight test = calibration trigger
H8	Bond work index	12.54 kWh/t	[ref.]; CR.5107's $W = 3.12$ kWh/t carries the Bond-validity caveat (§3.3)
H9	Planning rule	FeedLime demand drives; AgLime ≤ 135 kt cap; zero surplus	client decision 2026-08-12
H10	AgLime acceptance criterion	$\geq 95\% < 1.7$ mm	client-adopted 2026-08-12 (achieved: 98.8 %)

Open points for the reviewer's attention: (i) the CR.5107 power figure (Bond far out of its comfort zone at ratio 18 — the least certain number of the document); (ii) the 1.7 mm Karra validity limit; (iii) vendor gaps: SR.5105/SR.5115 installed areas, CR.5107 motor, BF.5101 datasheet; (iv) the $\approx 16\,400$ t FeedLime seasonal storage requirement; (v) the imperfection arbitration (0.15 vs 0.10).

6 • References

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Provenance — deterministic engine, git commit 69c40e8, models M3 (Karra partition), M4 (screen area), M5 (JKMRRC impact), M1 (Rosin–Rammmler product), M2 (Bond power), run 2026-08-12, 110 regression tests passed. Weekly-plan arithmetic computed in the stamped verification script from the engine's planning module (``run_required_hours``: 2 729.5 dry + 241.2 rain effective hours). Every figure is engine output. Document typeset with KaTeX.